

6. Adaptive Thresholding

Pixel values: [60, 80, 120, 140]

(a) Interpretation of the segmentation challenge

A single global threshold assumes uniform illumination across the whole image. When illumination varies (shadows, uneven lighting), the same global threshold can misclassify pixels in darker or brighter regions, since an intensity that means "object" in one region may mean "background" in another.

(b) Justification for adaptive thresholding

Adaptive (local) thresholding computes a separate threshold for each local neighbourhood, typically based on the local mean or a statistic of that region, so the segmentation adapts to local lighting conditions instead of using one fixed value for the entire image.

(c) Apply local threshold (mean = 100)

Rule: assign 0 if pixel < local mean (100), assign 1 if pixel $\geq$ local mean (100).

(d) Computed binary output

Pixel value	60	80	120	140
Binary output	0	0	1	1

(e) Interpretation of result

Pixels 60 and 80 (below the local mean) are grouped as background (0); pixels 120 and 140 (above the local mean) are grouped as foreground (1) — the local mean successfully separates the two intensity clusters present in this neighbourhood.